Early Detection and Intervention of a High-risk Neonate

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Introduction

High-risk neonate is any neonate, regardless of birth weight, size, or gestational age, who has a greater than average chance of morbidity or mortality, especially within the first 28 days of life. Children are identified as being at high risk for developmental difficulties if they experience problems before, during, or after birth that are known to be associated with later difficulties in development.

TABLE 1: Assigning the risk of neurodevelopment disability based on perinatal risk factors.

Risk factor	*Mild*	*Moderate*	*Severe*
Gestation (in weeks)	34–36	29–33	≤28
Birth weight (in grams)	1,501–2,000	1,001–1,500	≤1,000
Stage of HIE	HIE–1 (Levine classification)	HIE–2/chest compressions/ medications	HIE–3/abnormal neuro examination at 2 weeks
Hypoglycemia (mg/dL)	☑ <50 after day 3 ☑ <40 on day 1–day 2	Need for IV dextrose	Seizure/apnea/persistent jitteriness
Neonatal jaundice	Bilirubin ≥ 20 mg/dL	Exchange transfusion	Encephalopathy/kernicterus
Infection	Clinical sepsis requiring IV antibiotics	Culture positive sepsis	Meningitis/shock/ coagulopathy
Seizures	Hypocalcemia	Requiring AED	Abnormal EEG or neuro-logical examination at 2 weeks
Circulation		PDA requiring treatment (in ≤28 weeks)	Shock requiring inotropes

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Risk factor	*Mild*	*Moderate*	*Severe*
Ventilation	Oxygen/HFNC/CPAP	Needs invasive ventilation	Respiratory support at 28 days or invasive ventilation for 7 days
Place of birth		Outborn	Unsafe transport
Antenatal	☑ Gestational diabetes ☑ Twins ☑ PIH ☑ Previous IUD/stillbirth	☑ Eclampsia ☑ Abruptio placenta ☑ Absent/reversal of end-diastolic flow ☑ Category C fetal heart tracing ☑ Less than third centile fetal growth	IU infection
Neuroimaging	IVH–1	IVH–2	☑ PVL–2 or more ☑ IVH–3 or more ☑ Basal ganglia hyperintensity in HIE ☑ Loss of PLIC in HIE

(AED: antiepileptic drug; CPAP: continuous positive airway pressure; EEG: electroencephalogram; HFNC: high-flow nasal cannula; HIE: hypoxic-ischemic encephalopathy; IU: intrauterine; IUD: intrauterine death; IV: intravenous; IVH: intraventricular hemorrhage; PDA: patent ductus arteriosus; PIH: pregnancy-induced hypertension; PLIC: posterior limb of the internal capsule; PVL: periventricular leukomalacia)

TABLE 2: Assigning levels of care.

Risk of ND delay	*Low risk*	*Moderate risk*	*High-risk*
Aim of ND follow-up	Screening and stimulation	Screening and early intervention	Diagnosis and specific intervention
Responsibility	Parent/primary physician	Pediatricians	Developmental pediatrician/therapist
Intervention	Early stimulation	Early intervention	Definitive therapy
Place of therapy	In clinic/home	Developmental unit	Child developmental center
Roles	Pediatrician-habilitation		Rehabilitation

(ND: neonatal death)

TABLE 3: Checklist for high-risk neonate.

Checklist for high-risk neonate	***Age recommended (detailed to be followed as per National Consensus guidelines)***
Before discharge	☑ Assign risk ☑ Head circumference on WHO head centile chart ☑ Follow-up schedule with immunization visit
ROP screen	☑ At 4 weeks chronological age for all preterm (<34 weeks or < 2,000 g) Subsequently as per recommendation of ophthalmologist ☑ Visual acuity at 6 months and 1 year
Thyroid profile screen or 4-panel metabolic screen	Day 3–7
Neuro sonogram	First at day 3–5 and second at 40 weeks corrected age
OAE/BERA hearing screen	☑ OAE before discharge, if refer repeat at one and half month with first immunization ☑ BERA at 3 months

(BERA: brainstem evoked response audiometry; OAE: otoacoustic emission; ROP: retinopathy of prematurity)

The red flag signs as per the American Academy of Pediatrics (AAP) are presented in **Table 4**.

TABLE 4: Red flag signs as per the American Academy of Pediatrics (AAP).

At 9 months: Inability to	***At 18 months: Inability to***
1. Sit well with support	1. Stand and walk independently
2. Motor symmetry without established handedness	2. Should grasp and manipulate small objects
3. Should be able to grasp objects	3. Any sign of regression in GM/FM skills
4. Transfer objects from hand to hand	4. Drooling
5. Roll over to both sides	

(GM/FM: gross motor/function measure)

Red Flags for (Communication, Behavior, and Socialization)

If at 2 years, the mother is:

- ☑ Not satisfied with the child's speech or if there is regression in speech.
- ☑ Not satisfied with the child's behavior, he looks lost in his own world.
- ☑ If the child is not following more than two to three single line commands, or is hyperactive.
- ☑ If the child is having sensory issues such as hyperacusis, chewing difficulties, and difficulty holding crayons.
- ☑ If any of these are present individualized direct services should be started before 30 months of age of child.

Direct Observation Card for community-based screening and identification:

- ☑ Confirm that child can see and hear
- ☑ 2 months old infant—social smile should be achieved
- ☑ 4 months old infant—head holding
- ☑ 8 months old infant—should be sitting independently
- ☑ 12 months old infant—should be standing without support
- ☑ 15 months toddler—should speak Mama, Papa (specific) in his/her native language
- ☑ 18 months—follow one line command with no regression in communication, social, or play.
- ☑ If any of them is not achieved early intervention (EI) should be provided to the child.

TABLE 5: Amiel Tison angles.

Age	*0–3 months*	*3–6 months*	*6–9 months*	*9–12 months*
Adductor angle	40°–80°	70°–110°	120°–150°	130°–150°
Popliteal angle	80°–100°	90°–120°	110°–160°	150°–170°

Angle < : Hypertonia
Angle > : Hypotonia

Prechtl's General Movements Assessment

General movements (GMs) are part of the spontaneous movement repertoire and are present from early fetal life onward until the end of the first half a year of life. Prechtl's General Movements Assessment is simply a video of your baby lying on their back whilst they are awake, calm, and alert. The assessment can be completed from birth up to 20 weeks of age (corrected age). It is noninvasive and nondisruptive. This assessment method has shown its merit for the prenatal and postnatal evaluation of the integrity of the nervous system. It is the best predictor for cerebral palsy (CP). It is a reliable tool for identifying infants at risk for neuromotor deficits. Recent studies on preterm infants demonstrate that abnormal GMs also reflect impairments of brain areas involved in cognitive development.

Prechtl's General Movements Assessment (Fig. 1)

- ☑ Training for Prechtl's General Movements Assessment: https://general-movements-trust.info/69/dates/gt-basic-course
- ☑ Training for The Hammersmith Infant Neurological Examination (HINE): https://cre-auscpctn.centre.uq.edu.au/education/hine-training

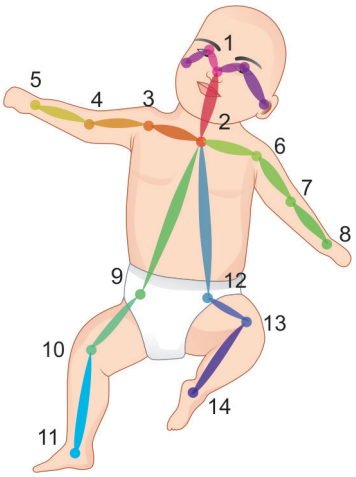

Fig. 1: Prechtl's general movements assessment.

The Hammersmith Infant Neurological Examination

- ☑ The Hammersmith Infant Neurological Examination (HINE) is a standardized and scoreable clinical neurological examination that can be used to assess infants from 2 to 24 months of age. The HINE contains 26 items across 5 domains and summing of scores in each domain provides a global score.
- ☑ More than 10 studies of "high-risk" newborns (preterm and term) have demonstrated that this assessment can be used in the diagnosis of CP. A HINE score <57 at 3 months 96% predictive of CP (sensitivity 96%; specificity 87%).
- ☑ It provides objective information about likely motor severity and distribution of CP. Scores below 40 predict nonambulant CP. It provides information on other aspects of neurological function other than motor. It has good interobserver reliability for all levels of clinical experience.
- ☑ Beside these the TDSC 1 and 2, Language Evaluation Scale Trivandrum (LEST), Trivandrum Autism Behavioral Checklist (TABC), M-CHAT R/F, BAYLEY 4, Denver Developmental Screening Tool (DDST), Gross Motor Function Classification (GMFC), Manual Ability Classification System (MACS), the Communication Function Classification System (CFCS), and the Eating and Drinking Ability Classification System (EDACS), can be used to identify the delay in specific domain.

The Hammersmith Infant Neurological Examination (FIG. 2)

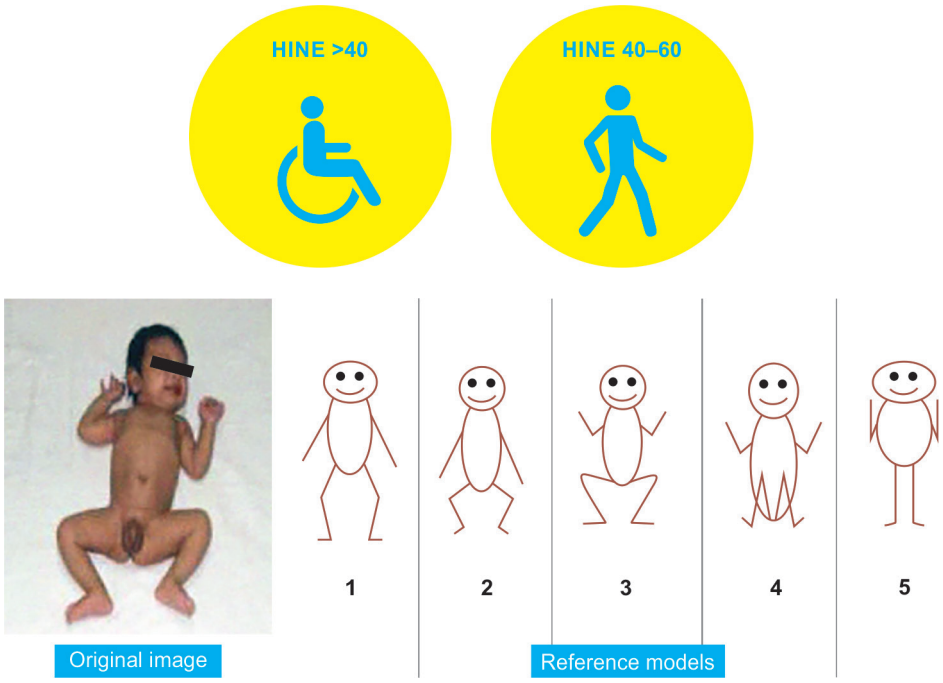

Fig. 2: The Hammersmith Infant Neurological Examination.

Regardless of the scale used, it is important to identify the problem list in developmental domains and to address each individually at the earliest and not to wait till a final diagnosis is made.

Multidisciplinary intervention is two or more healthcare disciplines work together toward a person centered, common goal, reducing the impact of disability and handicap, and help functional integration. Direct or definite services include occupational therapy, speech and language therapy, special education, physical therapy, behavioral modification, and other services such as assistive technology and nutrition may be required.

Management

First 1,000 days (antenatal 280 days + first year postnatal 365 days + second year postnatal 365 days) of life is scientifically proven to be the most significant and crucial period of an individual's life where EI should be targeted. Maximum synaptogenesis in brain takes place in this period.

Early intervention strategies should start early in the neonatal intensive care unit (NICU):

- ☑ Promote bonding through the Kangaroo method.
- ☑ Unnecessary noises of alarms should be reduced (1 hour noise-free period).
- ☑ Individual lighting units should be dimmed at night to simulate cycles of days.
- ☑ Passive exercises of the joints and gentle massage should be given to neonate.

Early intervention for head control:

- ☑ If the child does not lift his head, gently stroke downward over the neck muscle.
- ☑ Carry the child in upright position.
- ☑ Slowly lift the child from lying down position holding axilla, to sitting position and then slowly put him back to lying position thereby stimulating to lift and hold head (5 minutes × four times/day).

Early intervention for sitting:

- ☑ Make the child sit, provide support and toys, then slowly reduce support.
- ☑ Make her come to sitting position, hold her in this position and then make her lie down (5 minutes × four times a day).
- ☑ Make the child sit across his mother's knee, just tilt the child forward and sideways, so that he/she outstretches the hand. This helps in development of righting reflex.

Early intervention for standing:

- ☑ Make him come to standing position holding a stool and put toys.
- ☑ Help him in coming to sitting position on its own.
- ☑ Help him to walk with support holding both hands and later with one.
- ☑ Outdoor playing with children will help holistic development.

Early intervention for vision and socialization:

- ☑ The mother while maintaining eye to eye contact performs singing or talking keeping face 6–8 inch away.
- ☑ Hang brightly colored objects 12–15 inches above the crib.
- ☑ To promote visual following use multicolored small bulb series, which is to be hanged on the wall at the foot end of the child's bed/crib.
- ☑ *Promote symbolization:* It can be done by pointing and taking name of common objects when it comes in front of the child eyes.
- ☑ *To promote joint attention:* Sharing of something interesting which is at a distance by pointing it to the child.

Early intervention in speech and communication:

- ☑ Speak clearly and with eye-to-eye contact to the child.
- ☑ Quality communication between infant and family members and use single language.
- ☑ Make the child listen to different sound and do repetitions of what you say.
- ☑ Encourage him to share his experiences and to tell stories.
- ☑ No electronic gadgets before 2 years of life.

Early intervention for hand function:

- ☑ For opening of hand pull the thumb or fingers from base.
- ☑ Place an object in his hand and bend his fingers around it to promote hand grasp.
- ☑ Apply dough or colored clay on thumb and index finger to promote pincer grasp.

Further Reading

- ☑ CanChild Centre for Childhood Disability Research. Gross Motor Function Classification system—Expanded and Revised. [online] Available from https://www.canchild.ca/system/tenon/assets/attachments/000/000/058/original/GMFCS-ER_English.pdf. [Last accessed November, 2022].
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Injuries in Office Practice

Lead Author
Chetan Ginigeri

Co-Authors
Alok Bhandari, Ramakant Patil

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

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IAP Standard Treatment Guidelines Committee

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Remesh Kumar R

IAP Coordinator

Vineet Saxena

National Coordinators

SS Kamath, Vinod H Ratageri

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Krishna Mohan R, Vishnu Mohan PT

Members

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